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Tarea: Resuelva numericamente las ecuaciones de primer orden que se obtienen al aplicar MV para un MA(1) con ruido blanco normal y resolverlas numericamente con el método de Newton-Raphson.

MA(1): $y_t = \varepsilon_t + \beta_1 \varepsilon_{t-1}$, con $E[y_t | y_{t-1}] = \beta_1 \varepsilon_{t-1}$

$$L(\vec{\theta}) = \prod_{t=1}^{T} f_{Y_t | Y_{-1} = y_{t-1}}(y_t | y_{t-1})$$

$$l(\vec{\theta}) = \sum_{t=1}^{T} \log\left[\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} e^{-\frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}}\right] = \sum_{t=1}^{T}\left[\log\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} - \frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}\right]$$

$$= \sum_{t=1}^{T} -\frac{1}{2}\log(2\pi\sigma_\varepsilon^2) - \sum_{t=1}^{T}\frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}$$

$$l(\vec{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log(\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon}\sum_{t=1}^{T}(y_t - \beta_1\varepsilon_{t-1})^2$$

$$\frac{\partial L}{\partial \beta_1} = \frac{-1}{2\sigma_\varepsilon^2}\sum_{t=1}^{T}\left[2(y_t - \beta_1\varepsilon_{t-1})\frac{\partial \varepsilon_t}{\partial \beta_1}\right] = 0$$

CPO: $$\sum_{t=1}^{T}(y_t - \beta_1\varepsilon_{t-1})\frac{\partial \varepsilon_t}{\partial \beta_1} = 0$$

$$\frac{\partial \varepsilon_t}{\partial \beta_1} = \frac{\partial(y_t - \beta_1\varepsilon_{t-1})}{\partial \beta_1}, \quad \frac{\partial y_t}{\partial \beta_1} = 0$$

$$\boxed{\frac{\partial(-\beta_1\varepsilon_{t-1})}{\partial \beta_1} = -\varepsilon_{t-1} - \beta_1\frac{\partial \varepsilon_{t-1}}{\partial \beta_1} = \frac{\partial \varepsilon_t}{\partial \beta_1}}$$

Pero $\frac{\partial \varepsilon_{t-1}}{\partial \beta_1} = \frac{\partial(y_{t-1} - \beta_1\varepsilon_{t-2})}{\partial \beta_1} = -\varepsilon_{t-2} - \beta_1\frac{\partial \varepsilon_{t-2}}{\partial \beta_1}$ ...

Usamos N.R $\rightarrow$ $x_n = x_{n-1} - \frac{f(x_{n-1})}{F(x_{n-1})}$ $\rightarrow$ CPO $\rightarrow$ CSO

$$\frac{\partial CPO}{\partial \beta_1} = \sum_{t=1}^{T}\left[\frac{\partial \varepsilon_t}{\partial \beta_1}\cdot\frac{\partial \varepsilon_t}{\partial \beta_1} + \varepsilon_t\frac{\partial^2 \varepsilon_t}{\partial \beta_1^2}\right]$$

$$= \sum_{t=1}^{T}\left[\left(\frac{\partial \varepsilon_t}{\partial \beta_1}\right)^2 + \varepsilon_t\frac{\partial^2 \varepsilon_t}{\partial \beta_1^2}\right] \rightarrow CSO$$

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$$\beta_1^{(1)} = \beta_1^{(0)} - \frac{\sum_{t=1}^{T} \varepsilon_t \cdot \frac{\partial \varepsilon_t}{\partial \beta_1}}{\sum_{t=1}^{T} \left[\left(\frac{\partial \varepsilon_t}{\partial \beta_1} \right)^2 + \varepsilon_t \frac{\partial^2 \varepsilon_t}{\partial \beta_1^2} \right]}$$

→ Con la CSO y la CPO evaluada en $\beta_1^{(0)}$

→ ¡Python!